

Programming Methodology - SE1012

Lab 3: Introduction to C Programming



Using Linux Machines

Access Lab 3 via GitHub:

<https://classroom.github.com/a/-P0VeEcy>

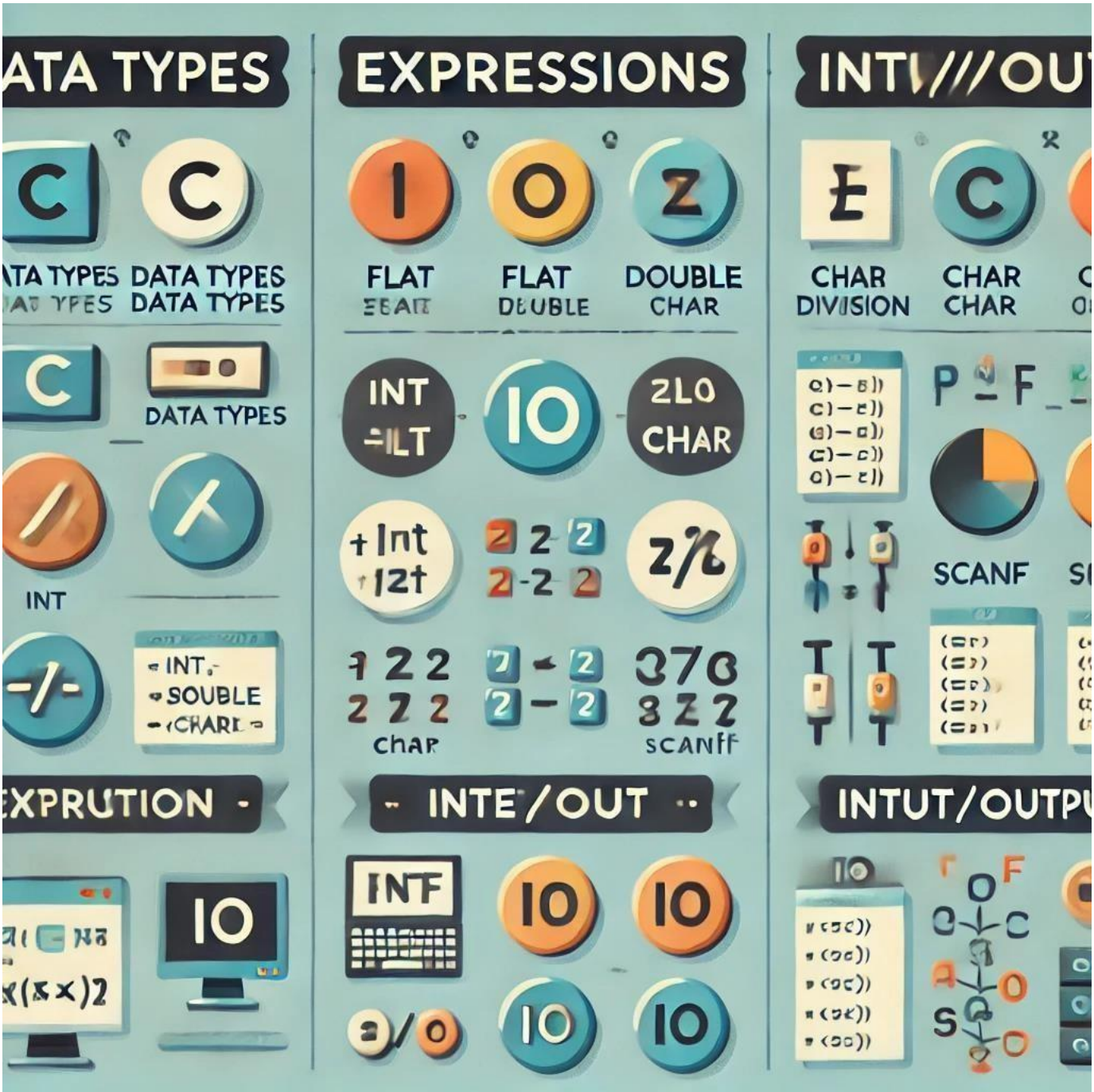
Clone the repository

Outline of Today's Lab



- Data Types / Expressions / IO
- Activity 1: Design of a Hydroelectric Dam
- Activity 2: A Jet Fighter Launched from an Aircraft-Carrier

Data Types / Expressions / IO



Data Types

Type	Storage Size	Range	Precision
char	1 byte	-128 to 127 or 0 to 255	
unsigned char	1 byte	0 to 255	
signed char	1 byte	-128 to 127	
int	2 or 4 bytes	-32,768 to 32,767 or -2,147,483,648 to 2,147,483,647	
unsigned int	2 or 4 bytes	0 to 65,535 or 0 to 4,294,967,295	
short	2 bytes	-32,768 to 32,767	
unsigned short	2 bytes	0 to 65,535	
long	8 bytes	-9223372036854775808 to 9223372036854775807	
unsigned long	8 bytes	0 to 18446744073709551615	
float	4 bytes	1.2E-38 to 3.4E+38	6 decimal places
double	8 bytes	2.3E-308 to 1.7E+308	15 decimal places
long double	10 bytes	3.4E-4932 to 1.1E+4932	19 decimal places

Expressions

Arithmetic Operator	Meaning	Examples
+	addition	<code>5 + 2</code> is <code>7</code>
		<code>5.0 + 2.0</code> is <code>7.0</code>
-	subtraction	<code>5 - 2</code> is <code>3</code>
		<code>5.0 - 2.0</code> is <code>3.0</code>
*	multiplication	<code>5 * 2</code> is <code>10</code>
		<code>5.0 * 2.0</code> is <code>10.0</code>
/	division	<code>5.0 / 2.0</code> is <code>2.5</code>
		<code>5 / 2</code> is <code>2</code>
%	remainder	<code>5 % 2</code> is <code>1</code>

Input/Output Functions

`printf` Function

The `printf` function is used to print output to the console. It includes several components:

```
printf("That equals %f kilometers.\n", kms);
```

- **Function name:** `printf`
- **Function arguments:** Everything within the parentheses
- **Format string:** `"That equals %f kilometers.\n"`
- **Print list:** `kms`

`scanf` Function

The `scanf` function is used to read input from the console. The syntax and an example are provided below:

Syntax:

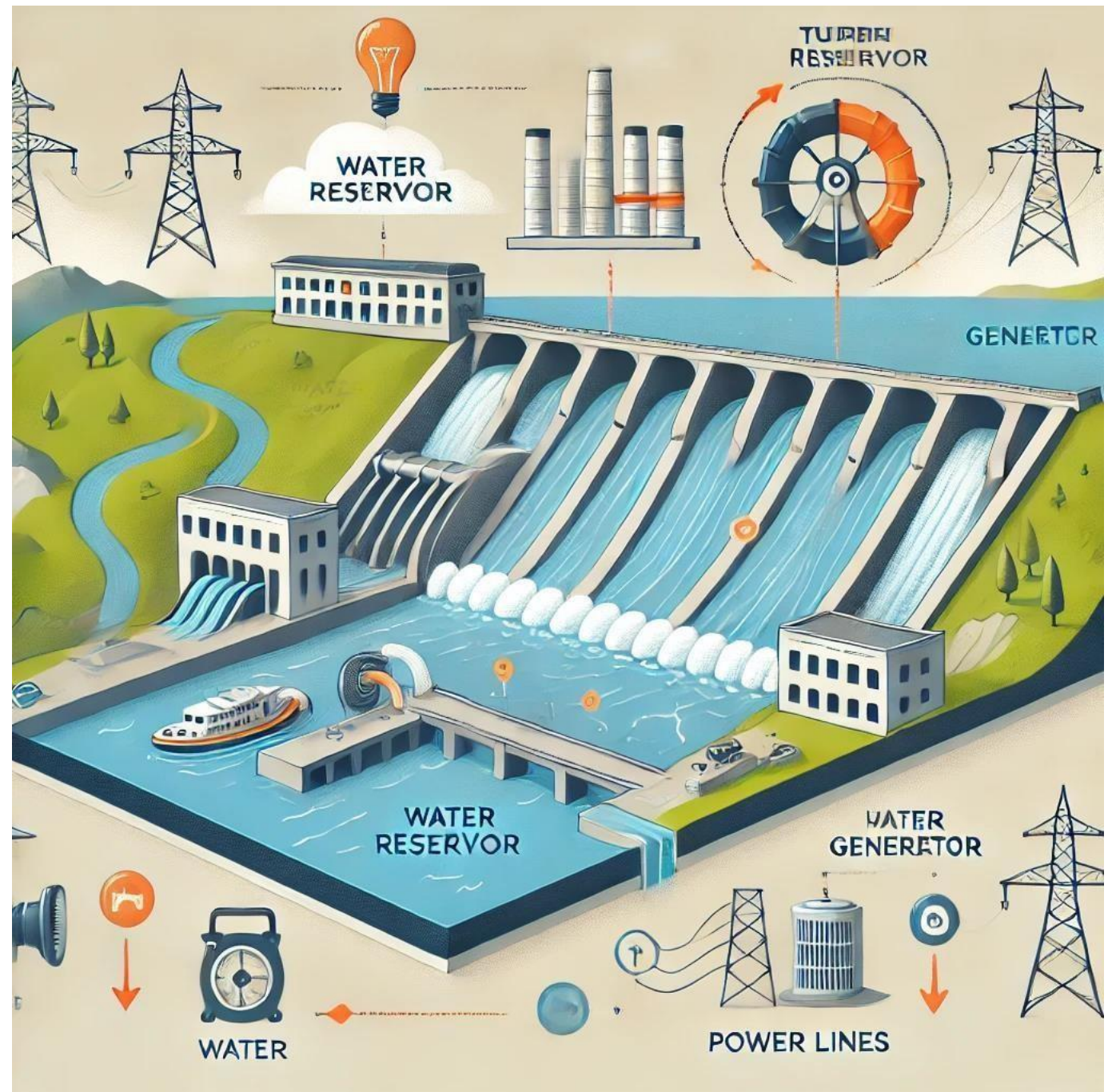
```
scanf(format string, input list);
```

Example:

```
scanf("%c%d", &first_initial, &age);
```

Placeholder	Variable Type	Function Use
<code>%c</code>	<code>char</code>	<code>printf</code> / <code>scanf</code>
<code>%d</code>	<code>int</code>	<code>printf</code> / <code>scanf</code>
<code>%f</code>	<code>double</code>	<code>printf</code>
<code>%lf</code>	<code>double</code>	<code>scanf</code>
<code>%f</code>	<code>float</code>	<code>printf</code> / <code>scanf</code>

Activity 1: Design of a Hydroelectric Dam



Design of a Hydroelectric Dam

Part 3: Graded Exercises

Exercise 1: Write a C program to assist in the design of a hydroelectric dam. Prompt the user for the height of the dam and for the number of cubic meters of water that are projected to flow from the top to the bottom of the dam each second. Calculate how many megawatts (1MW = 10^6 W) of power will be produced if 90% of the work done on the water by gravity is converted to electrical energy.

Note: The mass of one cubic meter of water is 1000 kg. Use 9.80 meters/second² as the gravitational constant `g`. Be sure to use meaningful names for both the gravitational constant and the 90% efficiency constant. For one run, use a height of 170 m and a flow of 1.30×10^3 m³/s.

Formulas:

```
Power =  $\eta$  * Work  
Work = m * g * h
```

- `$\eta$` is the efficiency of the conversion (in this case, 90%, so $\eta=0.9$).
- `m` is the mass of water flowing per second.
- `g` is the gravitational constant (9.80 m/s²).
- `h` is the height of the dam.

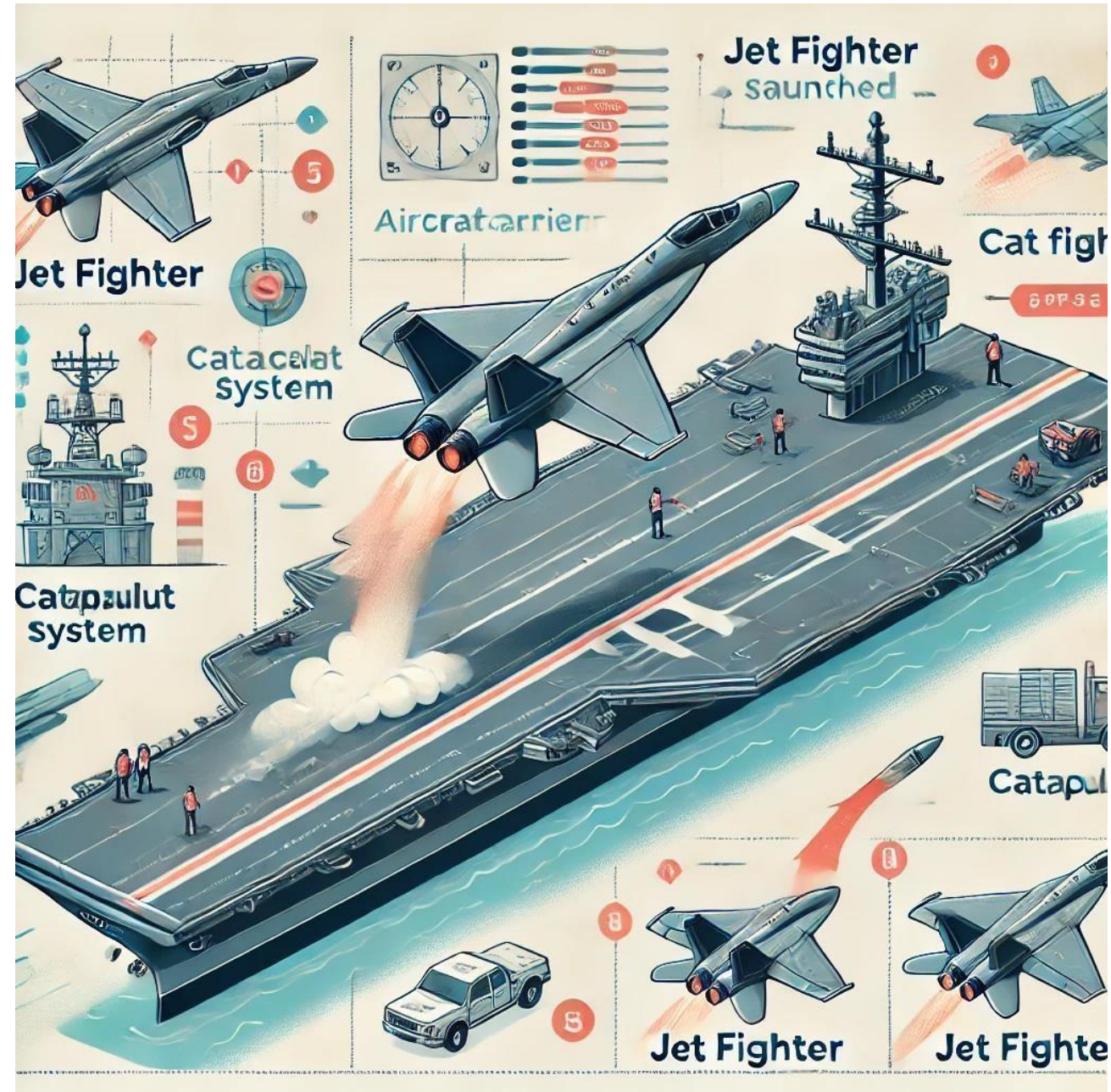
Test Case 1:

- Input: Height of the dam: 170 meters, Flow rate of water: 1.30×10^3 cubic meters per second.
- Output: Predicted power generated by the hydroelectric dam: 1949.22 MW

Test Case 2:

- Input: Height of the dam: 100 meters, Flow rate of water: 800 cubic meters per second.
- Output: Predicted power generated by the hydroelectric dam: 705.6 MW

Activity 2: A Jet Fighter Launched from an Aircraft-Carrier



A Jet Fighter Launched from an Aircraft-Carrier

Exercise 2: Write a program that calculates the acceleration (m/s^2) of a jet fighter launched from an aircraft-carrier catapult, given the jet's takeoff speed in km/hr and the distance (meters) over which the catapult accelerates the jet from rest to takeoff. Assume constant acceleration. Also calculate the time (seconds) for the fighter to be accelerated to takeoff speed. When you prompt the user, be sure to indicate the units for each input.

For one run, use a takeoff speed of 278 km/hr and 94 meters. Relevant formulas:

$$v = at$$
$$s = \frac{1}{2} at^2$$



Lab Submission Guidelines

- Submit your codes to the GitHub Classroom

